

Exam – 1 Review

Concepts, formulas, mini-examples, and common traps

How to Think on this Exam

Model first, compute second. Sketch a micro-diagram, label what's given, write the right formula, then substitute carefully with units & signs. Keep exact values ($\sqrt{}$, π) until the final step unless the problem explicitly asks for a decimal.

1. Geometry on a Line & in the Plane

Core Facts

Betweenness (Segment Addition): If B lies between A and C on a line, then $AB + BC = AC$.

Distance (1D): between x_1, x_2 is $|x_2 - x_1|$. **Distance (2D):** $d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Fraction along a segment (number line): If P is a fraction f of the way from A to B , $P = A + f(B - A)$ ($0 \leq f \leq 1$).

Partition by ratio $m : n$ (directed $A \rightarrow B$): $P = A + \frac{m}{m+n}(B - A)$.

Midpoint: $M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$. **Missing endpoint:** solve midpoint equations for the unknown endpoint.

Mini-example (ratio on a line). $A = -6$, $B = 10$. Point P is $\frac{3}{4}$ from A toward B .

$$P = -6 + \frac{3}{4}(10 - (-6)) = -6 + \frac{3}{4} \cdot 16 = -6 + 12 = 6.$$

Common traps:

- Direction matters on a number line: if C is left of A , $AC = |C - A|$ still.
- In 2D distance, $(x_2 - x_1)^2$ and $(y_2 - y_1)^2$ are *squared* (signs vanish after squaring).

2. Angle Basics

Core Facts

Straight angle = 180° , **Right** = 90° . **Vertical angles** are congruent. **Linear pair** $\Rightarrow$ supplementary.

Angle bisector splits an angle into two equal angles. **Complementary** sum to 90° .

Supplementary sum to 180° .

Mini-example. If $\angle ABD = (x + 20)^\circ$ and $\angle DBC = (2x - 10)^\circ$ with $\overrightarrow{BD}$ a bisector, set them equal: $x + 20 = 2x - 10 \Rightarrow x = 30$. Then $\angle ABC = 2 \cdot 50^\circ = 100^\circ$.

Trap: “Vertical” means formed by two intersecting lines (not “stacked”). Vertical $\Rightarrow$ equal; linear pair $\Rightarrow$ add to 180° .

3. Coordinate Geometry: Perimeter & Area

Core Formulas

Rectangle $P = 2(\ell + w)$, $A = \ell w$; Triangle $A = \frac{1}{2}(\text{base})(\text{height})$.

Circle $C = 2\pi r$, $A = \pi r^2$. **Square from diagonal d :** $s = \frac{d}{\sqrt{2}}$, $A = s^2 = \frac{d^2}{2}$.

Right triangle check: If sides a, b, c with c largest satisfy $a^2 + b^2 = c^2$, triangle is right.

Mini-example. Square with opposite vertices $(1, 1)$ and $(5, 5)$: $d = \sqrt{(5-1)^2 + (5-1)^2} = 4\sqrt{2} \Rightarrow s = \frac{4\sqrt{2}}{\sqrt{2}} = 4$. Then $P = 16$, $A = 16$.

Traps:

- Use *vertical distance* (difference in y) for height above a horizontal base.
- For circles, if endpoints of a diameter are given, center = midpoint, radius = half the distance.

4. Rigid Motions (about the Origin)

Rules

Reflection in x -axis: $(x, y) \mapsto (x, -y)$. Translation by (a, b) : $(x, y) \mapsto (x + a, y + b)$.

Rotation 90° CCW: $(x, y) \mapsto (-y, x)$. Rotation 180° CCW: $(x, y) \mapsto (-x, -y)$.

Rotation 270° CCW: $(x, y) \mapsto (y, -x)$.

Invariants: distances, perimeters, areas, angle measures, and parallelism are preserved.

Mini-example (order matters). Translate $(-2, 3)$ by $(5, -4) \rightarrow (3, -1)$, then rotate 90° CCW $\rightarrow (1, 3)$. If you rotate first $\rightarrow (-3, -2)$, then translate $\rightarrow (2, -6)$. Not the same!

5. 3D Models: Surface Area & Volume

Must-Know Formulas

Rectangular prism: $V = \ell wh$, $SA = 2(\ell w + \ell h + wh)$.

Right circular cylinder: $V = \pi r^2 h$, $SA_{\text{total}} = 2\pi r(h + r)$ (lateral = $2\pi rh$).

Right circular cone: $V = \frac{1}{3}\pi r^2 h$, $SA_{\text{total}} = \pi r(r + s)$, where $s = \sqrt{r^2 + h^2}$.

Right square pyramid: $V = \frac{1}{3}a^2 h$, $SA_{\text{total}} = a^2 + 2as$, where $s = \sqrt{h^2 + (a/2)^2}$.

Sphere: $V = \frac{4}{3}\pi r^3$, $SA = 4\pi r^2$.

Modeling tips:

- Decide if you need *slant height* s (cones, pyramids) or *vertical height* h .
- For composite solids, *don't double count* interior interfaces on surface area.
- “Same volume” comparisons often cancel π and isolate the key variable quickly.

6. Measurement: Precision, Accuracy, Rounding, Significant Figures

Concepts & Rules

Precision = repeatability (small spread). **Accuracy** = closeness to true value.

Absolute error = $|\text{measured} - \text{true}|$. **Relative error** = $\frac{\text{abs. error}}{\text{true}}$. **Percent error** = $\text{relative} \times 100\%$.

Rounding intervals: If a reported value v is rounded to the nearest u (unit, tenth, hundredth, ...), then

$$\text{true} \in \left[v - \frac{u}{2}, v + \frac{u}{2} \right).$$

Propagate ranges through sums/products to *bound* true results.

Significant figures — counting:

- Nonzero digits: significant.
- Zeros between sig digits: significant.
- Leading zeros: not significant.
- Trailing zeros: significant if a decimal point is shown; otherwise use scientific notation to clarify.

Operations rules:

- **Add/Subtract:** round to the least precise decimal place among inputs.
 - **Multiply/Divide:** round to the least number of significant figures among inputs.
- Best practice:** keep guard digits in work; round *once* at the final answer (unless instructed).

Mini-examples.

- *Rounding interval:* 7.84 m to nearest 0.01 $\Rightarrow [7.835, 7.845)$ m.
- *Add/sub sig figs:* $12.07 + 0.3 + 0.025 = 12.395 \rightarrow 12.4$ (least precise is tenths).
- *Mult/div sig figs:* $3.20 \times 0.054 = 0.1728 \rightarrow 0.17 = 1.7 \times 10^{-1}$ (2 s.f.).

Traps:

- Don't round intermediate steps aggressively—save rounding for the end.
- For area/volume with rounded inputs, give (i) a sig-fig result *and* (ii) a range by interval bounds.

7. Lightning Checks (try without a calculator first)

- 1) Number line: $A = -8$, $C = 7$. Point P is 70% of the way from A to C . Find P .
- 2) Angle: vertical angles are $(2y + 14)^\circ$ and $(5y - 13)^\circ$. Solve for y and the angle measure.
- 3) Rigid motion: rotate $(x, y) = (4, -1)$ by 270° CCW about the origin.
- 4) 3D: a cone has $r = 6$ and slant $s = 10$. Find h and V .
- 5) Sig figs: compute $(7.16 - 3.2)/0.041$ using correct add/sub and then mult/div rules.

Answers (self-check)

1) $P = -8 + 0.7(15) = 2.5$. 2) $2y + 14 = 5y - 13 \Rightarrow y = 9$, angle = 32° . 3) $(y, -x) = (-1, -4)$. 4) $h = \sqrt{10^2 - 6^2} = 8$, $V = \frac{1}{3}\pi \cdot 36 \cdot 8 = 96\pi$. 5) $7.16 - 3.2 = 3.96 \rightarrow 4.0$; $4.0/0.041 \approx 97.6 \rightarrow 98$ (2 s.f.).

Pro tip: Before submitting, ask: “Did I choose the right model? Did I label units? Did I round at the very end with the correct rule?”